

DATA266 Lab 1

Team 5

Sneha Singh

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Abstract

This report is Team 5's write-up for DATA266 Lab 1. Sneha Singh and Ritika Mukesh Neema each trained a character-level GPT on TinyStories, three Yelp polarity classifiers, and a CycleGAN for photo-to-Monet transfer. Each person used their own architecture and their own hyperparameters. Each part below gives one person's run, then the other person's run, then a comparison of the two. Kaggle scores are still open.

The repository is https://github.com/snehas-SJSU/Data_266-Lab1.

Introduction

Each part below has two individual write-ups and then one team comparison. Section A is Sneha's model, metrics, failures, and hardware. Section B is Ritika's. Section C puts the two runs side by side. Code for Sneha lives under `sneha_singh/`. Code for Ritika lives under `ritika_mukesh_neema/`. Raw TinyStories and the Monet and photo folders are linked from the README because they are too large for git. Yelp is downloaded by the notebook from Hugging Face.

What each of us built

Part	Sneha Singh (her own model)	Ritika Mukesh Neema (her own model)
1 — GPT (TinyStories)	4 layers, emb 256, context 128, 10 epochs, RTX 5090	4 layers, emb 128, context 256, 12 epochs, Colab Tesla T4
2 — Yelp polarity	BiLSTM best, macro-F1 0.942, full train set, Colab Tesla T4	BiLSTM best, macro-F1 0.934, full 560k set, Colab Tesla T4
3 — CycleGAN	Batch 4, Colab Tesla T4, about 4.3 h, FID 89.99	Batch 1, RTX 5090, about 28 min, FID 123.70

We share the raw data. The rows above are two separate runs of the same part, not a split of who had to do which part.

Part 1 — GPT from scratch

1.A Sneha Singh — individual

I trained a character-level GPT on TinyStories. The attention code is mine. I did not use `nn.Transformer` OR `nn.MultiheadAttention`.

The tokenizer is a simple `char_to_idx / idx_to_char` map, not BPE. The model has 4 layers, 4 heads, embedding size 256, and a context of 128 characters. A causal mask stops each position from seeing the future. I tied the token weights the usual way and trained with AdamW. Bias and LayerNorm do not get weight decay.

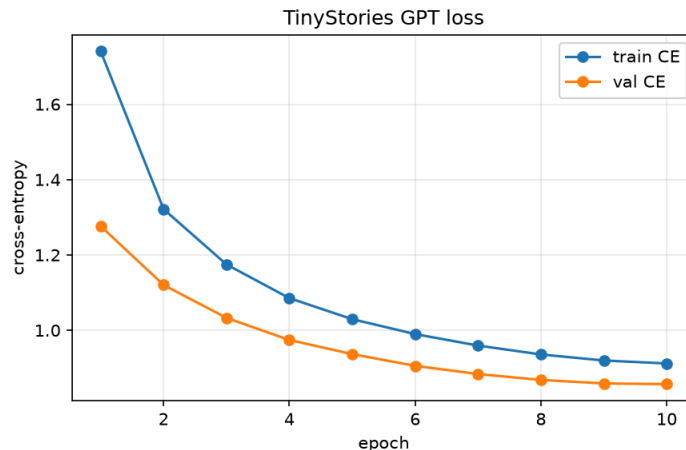
The reported run is the full one, not the old 2-epoch smoke test. I took my own split from `TinyStories-train.txt`: 100,000 train windows and 10,000 validation windows, seed 670. That text file is 1.79 GB, over GitHub's 100 MB limit, so it is not in git: https://drive.google.com/drive/folders/12MFVOo6T3QRW3X6THiDkh5svtNgL13W?usp=share_link

I trained for 10 epochs on an NVIDIA GeForce RTX 5090 with mixed precision. Learning rate was 0.001, with 100 warmup steps and then cosine decay. Batch size was 32. The run took 457 seconds and peaked at about 349 MB. There were no NaN losses. The checkpoint best.pt is 12 MB, so it is on git.

Checkpoint id: task1_llm/sneha_singh/checkpoints/best.pt. Log: reproducibility/raw_logs/sneha_singh/task1_llm/train_full.log.

Metric	Value
Train cross-entropy	0.912
Val cross-entropy	0.857
Perplexity	2.36
Bits per character	1.236
Generalization gap (val – train)	-0.055
Top-1 next-character accuracy	0.729
Distinct-1 / 2 / 3	0.225 / 0.655 / 0.856
Repeated 4-gram rate	0.120
Max gradient norm / NaN count	5.42 / 0
Parameters	3.25M
Train tokens/sec	280,325
Generation tokens/sec	409
Peak memory / train time	349 MB / 457 s

The result I care about on this model is that the samples stay varied while the loss stays stable. Distinct-1 is 0.225, distinct-2 is 0.655, and distinct-3 is 0.856. The repeated 4-gram rate is 0.120. Next-character top-1 accuracy is 0.729. Training had no NaNs, peaked at about 349 MB, and ran at about 280,000 tokens per second. The loss falls smoothly and the validation curve stays close to the training curve. That is what I wanted from 10 epochs on this size of model.



Part 1 training and validation loss

The prompt I used for samples was “some changes. She added some nice colors”.

Failure cases

- **Repetition.** Greedy decoding loops: “You are very happy. You are very happy.”

- **Broken word.** The same decode prints “them them”. A character model does not know where a word ends.
- **Story drift.** Temperature 0.8 moves from colors to a box, a door, and a tree. The model only sees 128 characters at a time.

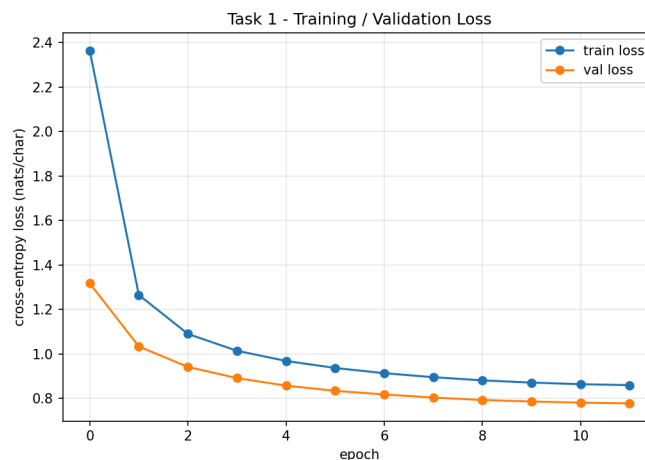
Full cases: [task1_llm/sneha_singh/failure_analysis.md](#)

1.B Ritika Mukesh Neema — individual

I wrote the attention myself. I did not use `nn.Transformer` or `nn.MultiheadAttention`. My model has 4 layers, 4 heads, embedding size 128, and a context of 256 characters. The tokenizer is character-level, built only from my training text, with a vocab of 221. I drew my own 100,000 / 10,000 windows from the full TinyStories train file, seed 6638. The checkpoint is `task1_llm/ritika_mukesh_neema/checkpoints/ckpt_final.pt`.

I trained for 12 epochs, batch 64, learning rate 0.0003, with linear warmup and cosine decay. Weight decay was 0.01 and gradients were clipped at 1.0. The run was on a Colab Tesla T4 and took 1,727 seconds, peaking at 1,749 MB. There were no NaN losses.

Metric	Train	Val
Cross-entropy	0.860	0.778
Perplexity	2.36	2.18
Bits per character	1.240	1.122
Generalization gap		-0.082
Top-1 next-character accuracy		0.755
Distinct-1 / 2 / 3		0.008 / 0.054 / 0.137
Repeated 4-gram rate		0.206
Max gradient norm / NaN count		4.99 / 0
Parameters		0.88M
Train tokens/sec		177,917
Generation tokens/sec		239
Peak memory / train time		1,749 MB / 1,727 s



Part 1 training and validation loss

Failure cases

- **Repetition.** All 10 greedy samples are the same story, and it loops: “You are very happy. You are very happy.”
- **Broken word.** Temperature sampling writes “designt”, which is not English.
- **Story drift.** The object moves from a box to a red ball to a bird.
- **Does not stop.** Generation does not stop at the story-boundary token, so a second story gets stuck on the end.

Full cases: [task1_llm/ritika_mukesh_neema/failure_analysis.md](#)

1.C Team comparison

	Sneha	Ritika
Architecture	4 layers, 4 heads, emb 256, context 128, char tokenizer, causal mask	4 layers, 4 heads, emb 128, context 256, char tokenizer, causal mask
Hyperparameters	10 epochs, batch 32, lr 0.001, warmup 100, seed 670, AdamW	12 epochs, batch 64, lr 0.0003, warmup + cosine, seed 6638, clip 1.0
Train / val CE	0.912 / 0.857	0.860 / 0.778
Perplexity / bits per char	2.36 / 1.236	2.18 / 1.122
Gen. gap / top-1 acc	-0.055 / 0.729	-0.082 / 0.755
Distinct-1/2/3	0.225 / 0.655 / 0.856	0.008 / 0.054 / 0.137
Repeated 4-gram / NaNs	0.120 / 0	0.206 / 0
Max gradient norm	5.42	4.99
Params / time / peak MB	3.25M / 457 s / 349	0.88M / 1,727 s / 1,749
Train / gen tokens per sec	280,325 / 409	177,917 / 239
Checkpoint	best.pt	ckpt_final.pt
Log	train_full.log	train_full.log

Strength. Ritika’s validation loss is 0.778, and Sneha’s is 0.857. Sneha’s distinct-1 is 0.225, and Ritika’s is 0.008. Each run leads on a different metric.

Weakness. Both greedy decoders loop on “You are very happy.”

Limitation. Embedding size, context length, and the machine differ, so the loss gap is not a pure architecture contest.

Next. Both of us would add a repetition penalty at decode time, and stop generation when the story-boundary token appears.

Part 2 — Yelp polarity

2.A Sneha Singh — individual

This is Yelp polarity, not IMDB. I learned the embeddings from scratch. I did not use a pretrained language model.

The Yelp files are not on Drive and not in git. The notebook downloads fancyzhx/yelp_polarity from Hugging Face when the run starts, so there is nothing for me to upload.

I wanted one simple model and two that can use word order, so a strong baseline would be obvious if the fancier models did not beat it.

1. **Baseline.** Mean pool over the embeddings, then a linear layer.
2. **BiLSTM.** Reads the review in both directions, so negation and longer sentences have a chance.
3. **TextCNN.** Filters of width 3, 4, and 5, meant to catch short phrases like “not good”.

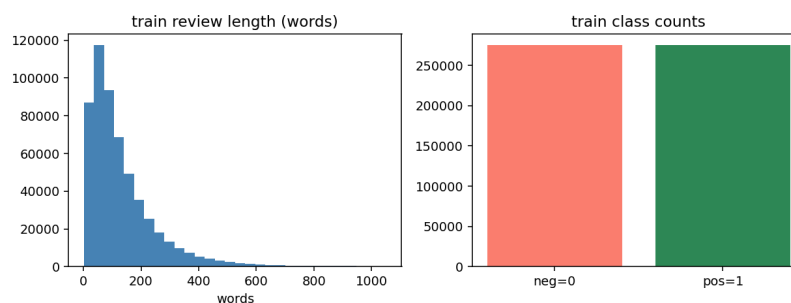
I lowercased, stripped punctuation, removed stopwords, and lemmatized. The vocabulary comes from the training text only. I held out 10,000 validation reviews from the 560,000-row train file, trained on 549,953 reviews, and tested on all 38,000 official test reviews. Five epochs, batch 64, embedding size 100, maximum length 128. All three models ran on a Colab Tesla T4. Peak memory was 228 MB, 344 MB, and 1,656 MB.

Checkpoints: task2_sentiment/sneha_singh/checkpoints/baseline_full.pt, experimental_a_full.pt, and experimental_b_full.pt. Logs: reproducibility/raw_logs/sneha_singh/task2_sentiment/baseline_full.log, experimental_a_full.log, and experimental_b_full.log.

Model	Acc	Macro-F1	ROC-AUC	MCC	Brier	ECE	Time
Baseline	0.932	0.932	0.979	0.863	0.052	0.006	233 s
BiLSTM	0.942	0.942	0.986	0.883	0.044	0.012	794 s
TextCNN	0.937	0.937	0.984	0.875	0.047	0.008	3646 s

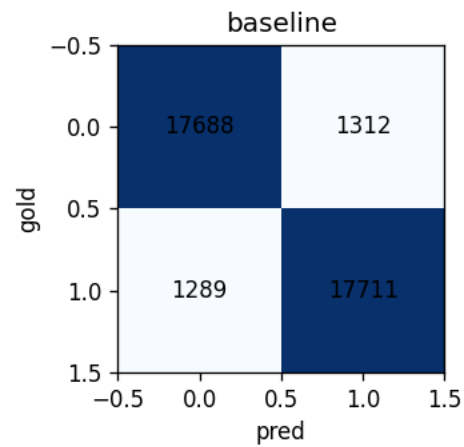
The BiLSTM has the best macro-F1. McNemar against the baseline is $p < 0.001$ for both the BiLSTM and the TextCNN, so on this 38k test set both beat the mean pool. Short reviews are a bit easier than long ones. Long-review macro-F1 is 0.927, 0.934, and 0.933 for baseline, BiLSTM, and TextCNN.

The length and class balance of this run:

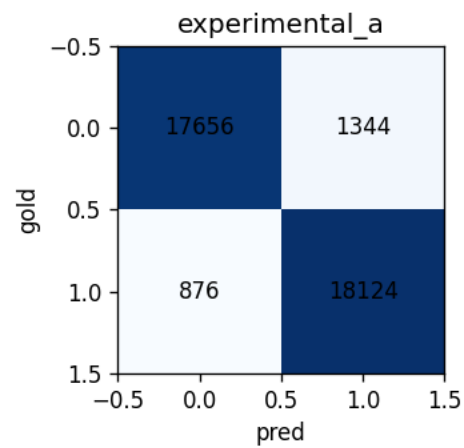


Yelp length and class balance

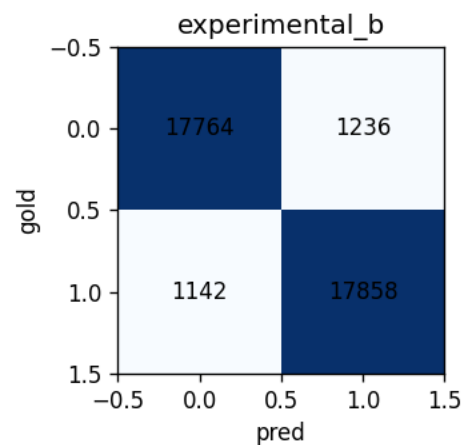
Confusion matrices on the test set:



Baseline confusion matrix



BiLSTM confusion matrix



TextCNN confusion matrix

Error review

I read 20 BiLSTM mistakes: five confident false positives, five confident false negatives, five near the decision threshold, and five long-review misses.

- **Confident false positive.** Gold negative, predicted positive: “wow love place everything clean new”.

- **Confident false negative.** Gold positive, predicted negative: “look know cox suck fact terrible business”.
- **Near the threshold.** “wow seems like taco bell arizona”.
- **Long review.** “given mixed review quite sure expect”.

The ones that show up most are praise words inside a negative review, a complaint that later turns around, and long reviews where the stars and the sentences pull apart. The fix I wrote down for these cases is to balance review length or train longer, and to add negation handling.

Full 20-row table: [task2_sentiment/sneha_singh/failure_analysis.md](#)

2.B Ritika Mukesh Neema — individual

I trained the same three families, with embeddings learned from scratch. My baseline is a mean pool plus a linear layer. My BiLSTM is there so negation can depend on word order. My TextCNN uses filter widths 3, 4, and 5 with global max-pool. I lowercased, stripped punctuation and HTML, removed stopwords, and used Porter stemming instead of lemmatization. I trained on the full Yelp polarity set, about 560,000 reviews, on a Colab Tesla T4. Checkpoints are `task2_sentiment/ritika_mukesh_neema/checkpoints/baseline.pt`, `bilstm.pt`, and `textcnn.pt`. I do not have confusion-matrix images in the repo. The numbers below are from my `metrics_report.csv`.

Model	Acc	Macro-F1	ROC-AUC	PR-AUC	MCC	Brier	ECE	Time	Peak MB
Baseline	0.922	0.922	0.974	0.973	0.843	0.059	0.007	125 s	2,123
BiLSTM	0.934	0.934	0.982	0.982	0.868	0.051	0.025	798 s	2,274
TextCNN	0.926	0.926	0.979	0.979	0.852	0.055	0.021	267 s	2,450

My BiLSTM is the best of my three. McNemar against my baseline is $p < 0.001$ for the BiLSTM and $p = 0.00022$ for the TextCNN, so on my test set both sequence models beat the mean pool.

Error review

I reviewed 20 TextCNN mistakes.

- **Confident false positive.** Sarcasm, with no negative words: “Is there REALLY even a Leonard...” A French review in the same set also fails, because the stemmer is English-only.
- **Confident false negative.** A positive review opens with “Cox sucks” before it turns around.
- **Near the threshold.** “The employees at this Target seemed unusually friendly...”
- **Long review.** “I used to love D&B... it has gone down hill.”

The fixes I wrote down are a language filter for non-English reviews, more weight on the words after “however”, and treating a score near 0.5 as a calibration margin rather than a hard error.

Full 20-case write-up: [task2_sentiment/ritika_mukesh_neema/failure_analysis.md](#)

2.C Team comparison

Same three model families. The tables use the same columns. Both of us trained on the full Yelp train file. Sneha tested on the official 38,000 reviews. Ritika tested on her own split of

that file.

Sneha Singh. 549,953 / 10,000 / 38,000, lemmatize, max length 128, Colab Tesla T4.

Checkpoints: baseline_full.pt, experimental_a_full.pt, experimental_b_full.pt.

Model	Acc	Macro-F1	ROC-AUC	PR-AUC	MCC	Brier	ECE	McNemar p	Params	Time	Peak MB
Baseline	0.932	0.932	0.979	0.979	0.863	0.052	0.006	—	10.68M	233 s	228
BiLSTM	0.942	0.942	0.986	0.986	0.883	0.044	0.012	< 0.001	10.76M	794 s	344
TextCNN	0.937	0.937	0.984	0.985	0.875	0.047	0.008	< 0.001	10.75M	3646 s	1,656

Ritika Mukesh Neema. Full 560k set, Porter stem, max length 200, Colab Tesla T4.

Checkpoints: baseline.pt, bilstm.pt, textcnn.pt.

Model	Acc	Macro-F1	ROC-AUC	PR-AUC	MCC	Brier	ECE	McNemar p	Params	Time	Peak MB
Baseline	0.922	0.922	0.974	0.973	0.843	0.059	0.007	—	3.00M	125 s	2,123
BiLSTM	0.934	0.934	0.982	0.982	0.868	0.051	0.025	< 0.001	3.24M	798 s	2,274
TextCNN	0.926	0.926	0.979	0.979	0.852	0.055	0.021	0.00022	3.12M	267 s	2,450

Precision, recall, the other F1 scores, bootstrap intervals, examples per second, and slice scores are in the next table. A slice cell is blank when that metrics file has no short-review or long-review column.

Metric	Sneha baseline	Sneha BiLSTM	Sneha TextCNN	Ritika baseline	Ritika BiLSTM	Ritika TextCNN
Precision macro	0.932	0.942	0.937	0.922	0.934	0.926
Recall macro	0.932	0.942	0.937	0.922	0.934	0.926
F1 micro	0.932	0.942	0.937	0.922	0.934	0.926
F1 weighted	0.932	0.942	0.937	0.922	0.934	0.926
Acc 95% CI	0.929–0.934	0.939–0.944	0.935–0.940	0.919–0.925	0.931–0.936	0.924–0.928
Macro-F1 95% CI	0.929–0.934	0.939–0.944	0.935–0.940	0.919–0.925	0.931–0.936	0.923–0.928
MCC 95% CI	0.859–0.868	0.879–0.888	0.870–0.879	0.838–0.849	0.863–0.873	0.847–0.857
Examples/sec	11,807	3,464	754	20,181	3,159	9,432
Short macro-F1	0.934	0.947	0.940			
Short error rate	0.065	0.052	0.059			
Long macro-F1	0.927	0.934	0.933			
Long error rate	0.072	0.065	0.066			

Strength. Sneha's BiLSTM macro-F1 is 0.942. Ritika's BiLSTM macro-F1 is 0.934. On both runs the BiLSTM beats that person's baseline, and McNemar says the lead is real.

Weakness. Both of us miss sarcasm, mixed reviews, and negation.

Limitation. The test splits are not the same cut of the file. Sneha lemmatizes and uses a maximum length of 128. Ritika stems and uses a maximum length of 200. The scores are not one shared test.

Next. Both of us would use the same maximum review length, so long reviews are scored the same way.

Part 3 — CycleGAN, photo to Monet

3.A Sneha Singh — individual

I trained an unpaired CycleGAN. The Kaggle direction is photo to Monet. The generators are mine. I did not use a pretrained model to make the images. Inception and the other nets in the eval script are only for measuring.

Each generator is a ResNet with 9 blocks at 256×256 , reflection padding, instance norm, and a tanh output. Upsampling is nearest-neighbor times two, then a stride-1 convolution, not a transposed convolution. Each discriminator is a PatchGAN trained with least-squares loss. The losses are adversarial, cycle L1 with $\lambda = 10$, and identity at half of that. Real labels are smoothed to 0.9. An image pool of 50 feeds the discriminator. Batch size is 4. I trained 40 epochs at a constant learning rate and 40 more with decay, and each epoch resamples about 800 photos. The shorter Monet loader is cycled so those photos actually get used. Scoring uses all 7,038 photos in the photo-to-Monet direction and all 300 Monet paintings in the other direction.

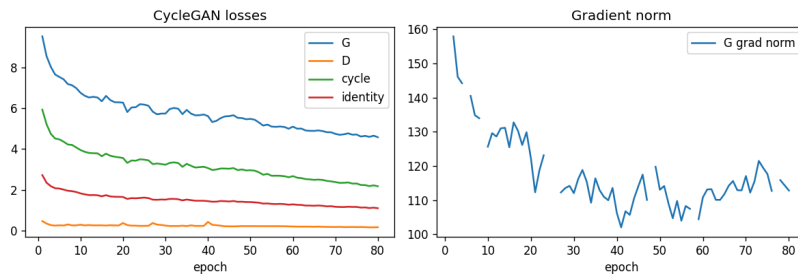
Training was on a Colab Tesla T4 with mixed precision. It took about 4.3 hours and peaked near 6568 MB. The log line is `device=cuda. best.pt` is 107.9 MB, over GitHub's 100 MB limit, so it is not in git: https://drive.google.com/drive/folders/12AgM95RbZAQUyouH7sukM9nurVTD3C9?usp=share_link

The Monet and photo folders are in one Drive folder: https://drive.google.com/drive/folders/1BXYfhW8uZ6umK1TZFW8Un62mZVK72L7Y?usp=share_link

Direction	FID	KID	MiFID	LPIPS	Content cosine	Cycle L1
Photo → Monet	89.99	0.018	0.403	0.452	0.532	0.112
Monet → photo	92.23	0.030	0.423	0.365	0.586	0.103

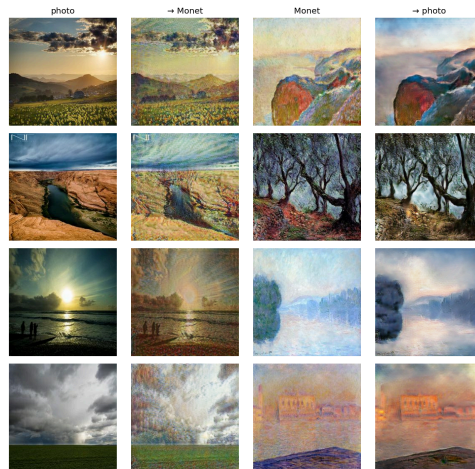
FID is the Fréchet distance between Inception features of the generated images and the real target images. MiFID here is the average cosine distance of those features after an equal subsample. Lower is better on both. My local combined score for the Kaggle direction is $(89.99 + 0.403) / 2 = 45.20$. The same two numbers are in `task3_gan/sneha_singh/submission.csv`. I have not uploaded that file yet, so public score, private score, and rank are still empty.

Losses fell across the 80 epochs. The discriminator loss ended near 0.17, which is low. The generator is not winning that fight, and I think that is why the brush texture gets noisy.



CycleGAN losses and gradient norm

The grid is four photos, the Monet version of each, four real Monets, and the photo version of those.



Photo, generated Monet, real Monet, generated photo

Human audit

What I see in that grid, and in the 30 images I audited:

- **Grain.** Skies and water pick up a repeated dab texture.
- **Color wash.** Some skies go flat, and sunsets brown out at the edges.
- **Soft reverse.** A few Monet-to-photo frames go soft or pick up a dark blob.
- **Content.** The layout of the photo usually survives. I did not mark any of my 30 as a copied training Monet.

On the 0–2 sheet my averages were style 1.4 and content 1.7, where higher is better, and artifacts 1.4, where higher is worse.

Full notes: [task3_gan/sneha_singh/failure_analysis.md](#)

Audit sheet: [task3_gan/sneha_singh/outputs/human_audit/audit_30.csv](#)

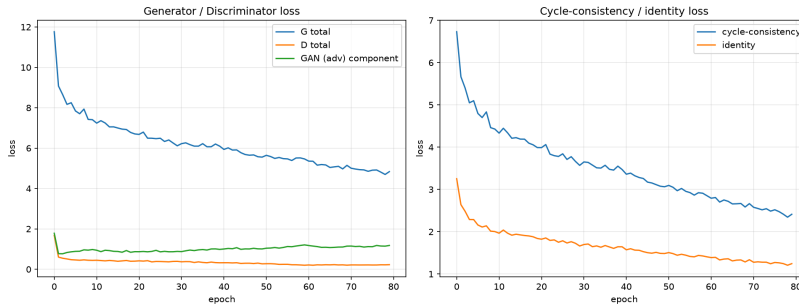
3.B Ritika Mukesh Neema — individual

I trained my own CycleGAN: two ResNet generators with 9 blocks, instance norm, and reflection padding, plus two 70×70 PatchGAN discriminators. I used least-squares adversarial loss, cycle L1 with $\lambda = 10$, identity loss with $\lambda = 5$, and an image pool of 50. Batch size was 1. I trained 40 epochs at a constant learning rate and 40 with decay, on an NVIDIA RTX 5090. The run took 1,660 seconds and peaked at 1,645 MB. There were no NaN losses. My photo-to-Monet direction is the one I label B2A. My photo-to-Monet FID is 123.70. Cycle L1, LPIPS, and content cosine are on 100 images, not the full photo set. ckpt_final.pt is

107.9 MB, so it is not in git: <https://drive.google.com/file/d/1Wlu22EKbp4QATijRbNBbDOFFGHLHUiNj/view>

Direction	FID	KID	Precision	Recall	Cycle L1	LPIPS	Content cosine
Photo → Monet (my B2A)	123.70	0.027	0.327	0.570	0.126	0.337	0.775
Monet → photo (my A2B)	120.72	0.040	0.613	0.207	0.109	0.405	0.873

Final generator loss was 4.84, discriminator loss 0.23, cycle loss 2.41, identity loss 1.24. Mean gradient norm was 57, max 315, NaN count 0. About 28.3 million parameters, 14.5 images/sec.



CycleGAN losses

My submission.csv records photo-to-Monet FID 123.70.

Failure notes

- **Cycle check.** Cycle L1, LPIPS, and content cosine are on 100 images, not the full photo set.
- **Human audit.** My audit folder is empty, and I do not have a failure_analysis.md for this run.

3.C Team comparison

Both models are ResNet-9 CycleGANs with a PatchGAN and least-squares loss. Lower FID is better. Checkpoint links are in sections 3.A and 3.B.

	Sneha photo → Monet	Sneha Monet → photo	Ritika photo → Monet	Ritika Monet → photo
Epochs / batch	40 + 40, batch 4	same run	40 + 40, batch 1	same run
FID	89.99	92.23	123.70	120.72
KID	0.018	0.030	0.027	0.040
Precision	0.540	0.660	0.327	0.613
Recall	0.293	0.089	0.570	0.207
Cycle L1	0.112	0.103	0.126	0.109
LPIPS	0.452	0.365	0.337	0.405
Content cosine	0.532	0.586	0.775	0.873
Params	28.3M	same run	28.3M	same run
Time / images per sec	15,342 s / 8.34	same run	1,660 s / 14.5	same run
Peak memory	6,568 MB	same run	1,645 MB	same run
NaN count	0	same run	0	same run

	Sneha photo → Monet	Sneha Monet → photo	Ritika photo → Monet	Ritika Monet → photo
Generator loss	4.574	same run	4.835	same run
Discriminator loss	0.168	same run	0.227	same run
Cycle loss	2.174	same run	2.412	same run
Identity loss	1.097	same run	1.244	same run
Gradient norm	112.82	same run	57.00 mean, 315 max	same run

Human audit score on the 30 photo → Monet images. Style and content are 0–2, higher is better. Artifacts are 0–2, higher is worse. Cohen’s kappa is the inter-rater agreement for both of us on the same 30 images. That column stays empty until Ritika scores them.

	Sneha	Ritika	Cohen’s kappa
Style	1.4		
Content	1.7		
Artifacts	1.4		

Strength. Sneha’s photo-to-Monet FID is 89.99. Ritika’s FID is 123.70. Ritika’s run is the faster one, about 28 minutes against about 4.3 hours.

Weakness. Both discriminator losses ended low, 0.17 for Sneha and 0.23 for Ritika. On Sneha’s images the brush texture is grainy.

Limitation. Batch size, training time, and the number of images in the cycle scores differ, so the two FID numbers are not a pure architecture contest.

Next. Both of us would train past 80 epochs, since the paper uses 200, and then upload one photo-to-Monet submission.csv.

Final comparison

Part	Sneha Singh	Ritika Mukesh Neema
1 — validation loss	4 layers, emb 256, context 128. 0.857	4 layers, emb 128, context 256. 0.778
1 — distinct-1	4 layers, emb 256, context 128. 0.225	4 layers, emb 128, context 256. 0.008
2 — best model	BiLSTM, 550k train, 38k test. Macro-F1 0.942	BiLSTM, full set. Macro-F1 0.934
3 — photo → Monet FID	Batch 4, Colab Tesla T4, about 4.3 h. 89.99	Batch 1, RTX 5090, about 28 min. 123.70

What is still open

Kaggle team PairProgramming_Team_5. One submission, from Sneha’s task3_gan/sneha_singh/submission.csv. The cells stay blank until the leaderboard numbers are pasted in.

	Public score	Private score	Rank
PairProgramming_Team_5			

The plots in this file are the png files already stored under each task folder. They are not copied again into report/.

Papers

1. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. 2017. <https://arxiv.org/abs/1706.03762>. The causal multi-head attention in Part 1 follows that design, written by hand.
2. Ronen Eldan and Yuanzhi Li. TinyStories: How Small Can Language Models Be and Still Speak Coherent English? 2023. <https://arxiv.org/abs/2305.07759>. Part 1 is trained on that dataset.
3. Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. 2017. <https://arxiv.org/abs/1703.10593>. Part 3 is that CycleGAN setup: two generators, two discriminators, cycle loss, and identity loss.